Engineering Intent

What apprenticeship might teach us about working with AI

An idea from technical work, explained through a simple example.



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RIFT asks whether agreeing on the standard, checking the work, and recording verified corrections can make AI-assisted projects more reliable.

This is a proposal to test. No RIFT performance results or academic validation are claimed.

What does “done” actually mean?

Imagine asking AI to turn inspection notes and photographs into a report. The report reads well. It looks complete. But one photograph is attached to the wrong piece of equipment.

The writing succeeded. The report still contains an error that matters.

Before the work starts

We could agree on a few checks: every photograph must match its equipment label, every statement must come from the notes, and missing information must stay marked as unknown.

Before calling it finished

Someone must compare the report with the original material. If a photograph is wrong, correcting that pairing is only part of the job. The rest of the report still needs checking.

This is the starting point for RIFT: make the meaning of “done” clear enough that a person can check it.

Fictional example. It uses no real inspection records and does not describe an observed RIFT result.

A correction needs a way to be checked

In technical apprenticeship, explaining a task and demonstrating competence are different parts of the work. A learner needs an opportunity to practice, receive useful feedback, and show that the correction carries into the next attempt.

Research on learning in context helps explain why instruction and practical activity belong together (Brown et al., 1989). That is the connection I want to explore.

The useful comparison

When working with AI, we can also explain the expected result, inspect what happened, and check another attempt. The comparison concerns how we organize the work.

The important difference

A person develops skills through lived experience. An AI model's training changes its internal numerical settings. Correcting an answer during a conversation does not automatically change those settings.

The next answer may improve because the correction is still in the conversation. A later session needs access to that lesson again, for example through a saved project record. Reflexion already studies feedback held in this kind of external memory (Shinn et al., 2023).

I am borrowing a practice from human instruction, not claiming that AI learns like a person.

RIFT makes the checking part of the job

1 Agree on the result

Use examples and questions to uncover what the person wants. Write down what must be correct, how it will be checked, and what is still uncertain.

2 Do the work and keep the evidence

Keep the source material and the first completed output. A claim that something was checked should point to an actual check.

3 Correct the failure and check again

Record what went wrong and what changed. Check the correction against the original standard. Also check that the change did not break something else. Set a limit on retries and involve a person when the next step is unclear.

4 Save a lesson only after review

A note saying “fixed” is not enough. Save the evidence and explain when the lesson applies. In the report example, future drafts would include an explicit photo-to-equipment check.

The person remains responsible for accepting the standard and the result. This develops my earlier working proposal (Hodgkin, 2026).

The contribution I am trying to make

I am not proposing that AI research has overlooked feedback or testing. Those ideas already have substantial research behind them.

What existing work contributes

GATE asks people questions to clarify their preferences (Li et al., 2025). Human–AI interaction guidelines address how people work with and correct AI systems (Amershi et al., 2019). CheckList tests specific model behaviors (Ribeiro et al., 2020).

Reflexion uses stored feedback between attempts (Shinn et al., 2023). AgentRx examines execution records to locate failures (Barke et al., 2026). Reliability research already draws on safety-critical engineering (Rabanser et al., 2026).

My proposed addition

I want to bring these practices together in a process that a skilled technician can use without first becoming a software engineer. Someone who knows how to inspect physical work may be able to apply that judgment to work produced with AI.

That is a hypothesis about making the process usable. The combination may help, or a simple checklist may do just as well. Calling it RIFT does not establish that it is new or better.

Would the extra checking be worth the effort?

Give people comparable fictional reporting tasks and compare four ways of working. Use the same AI model and tools, with the same overall limits on time and resources.

A A normal request, with ordinary follow-up.
B The request plus examples of the desired result.
C Examples plus agreed checks before starting.
D All of C, plus RIFT's recorded corrections and reviewed lessons from earlier practice tasks.

The main question

Does D produce more acceptable reports within the limit than C? This asks whether the added correction-and-memory process helps beyond agreeing on the checks.

How to keep the comparison fair

Decide the scoring rules before seeing the outputs. Have reviewers check reports without knowing which method produced them. Count planning and review time, not just AI generation time. Keep practice tasks separate from the scored tasks.

Repeat the comparison. One good report cannot show that a method works consistently; repeated-run evaluation is already part of agent research (Yao et al., 2025).

Score the first draft separately from the repaired report. A successful repair cannot erase an initial failure.

The process has to justify itself

The idea seems practical to build: a project brief, a short set of checks, a record of changes, and a person reviewing the result. It does not require retraining an AI model.

Practical to build does not mean useful. The process could take too long. People could agree on the wrong checks. A saved lesson could be mistaken or fit one task but not another.

What would count against it?

If the fuller process performs no better than the agreed-checks method, its additional steps have not earned their place. If it mainly helps experienced users, I cannot claim it makes the work easier for beginners. If the results vary widely, that uncertainty needs reporting.

The boundary matters

The first tests should use fictional records or isolated software tasks. They should not approve real equipment or replace qualified judgment. NIST's AI risk framework emphasizes context and accountability (Tabassi, 2023).

The question I want to contribute is whether the discipline used to teach and verify technical work can help people judge AI-produced work more consistently.

Independent research and personal views. No employer, client, or union endorsement is implied. AI assisted research, drafting, and layout. No experiment was run.

Sources and attribution

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Literature supports the cited concepts. It does not validate RIFT. Source review completed 25 October 2026.

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